

Part A

Scope

To control the scope of the project and ensure that our team delivers only the work required by AustStem, the team defined and set clear deliverables and developed them into the action register as well as into the work breakdown structure, covering software, hardware, and enclosures. To determine in-scope tasks and avoid scope creep in the project, the team met with AustStem and ruled out features such as the full Bee-haviour community platform. A thorough research of all AustSTEM-provided components was conducted, thus evaluating each against the project's scope to confirm suitability before committing to the final design. Responsibilities were then distributed as Jack and Ghad led hardware design, Ayrton developed schematics for the whole project, Ethan managed firmware libraries and Yunkai researched testing methodologies. Through structured delegations and utilising action registers with WBS to clarify the scope, the team is able to focus all its resources to meet AustStem demands.

Time

The team has to have the project ready for demonstration by Week 26. To achieve this, planning is essential in which Team 4 relied mainly on Gantt charts and critical path analysis to visualise the project schedule. The critical path highlighted how firmware and integration relied on the completion of hardware, BOM, and schematics, thus making the hardware phase the biggest driver of overall project length. Furthermore, the hardware phase was identified as a zero buffer zone and as a high risk area, in which any delays in that phase will impact the whole timeline of the project. In response, the team allowed a 2 week project buffer, as the internal deadline of the project is at week 24. Furthermore, the team adopted a Microsoft Solutions Framework (MSF) approach by distributing responsibilities, allowing all workstreams to run simultaneously and maintain progress. Finally, the 5 phases of Envisioning, Planning, Developing, Stabilising, and Deploying act as checkpoints that allow the team to continuously monitor milestone status allowing to catch and address errors before they stack up into large time consuming setbacks.

Risk

In the context of developing a monitoring system, the team anticipated a range of risks that are mainly attributed to working with a multi sensor embedded system. Addressing this, team 4 met with Mr Wayne Ireland identify and discuss potential hazards, producing a formal risk assessment (RSK2600305) that covered 4 key risks which were defective equipment, exposure to electricity, soldering iron use, and 3D printing. Upon our discussion about the risks levels and possible mitigations, it was deduced that the project does not impose any high risks rather it was mostly medium ratings with some of them being brought down to Low after consulting on specific control mitigation techniques. These included sourcing components from appropriate providers and only through the SECTE store, keeping the powering system within 6V, appropriately enclosing electrical connections and following approved safe work procedures for soldering and 3D printing. The team also agreed that before any material fabrication involving risks, we would check in with supervisors and complete any required training first. Moreover, it was raised by Mr Ireland that since the end users are school students, it is crucial to ensure that

all components are safely enclosed, which will be reviewed at each MSF phase to avoid any injury in a classroom environment.

Stakeholder

In this project there are 4 stakeholders that have an influence on the project. The main stakeholder is AustStem, as they are the project sponsor whose requirements and provided components have directly shaped the system's design. Specifically, Auststem specified the use of their Kookaberry, which the team had to incorporate as the foundational design constraint alongside the expectations of being low cost and educationally sustainable. End users are a very important stakeholder, being school students, as they influence all design decisions throughout the project, as every element from software to hardware must be intuitive, interactive, and simple for kids to operate without any prior technical knowledge. On the other hand, the ECTE 351 subject coordinators have set academic requirements that the team must satisfy while working on the physical product. These include documentation, presentations, and budget compliance as well as meeting specific subject outcomes this included having a GPS module into the final design. Finally, the team itself is an internal stakeholder, as being students working on a real engineering project for the first time, the learning curve each member faces can directly influence design decisions and the overall quality of the final product.

Part B

Integration

Integration management ensures that all subsystems of a project work together as a unified system. Moreover, integration arrangements sit at the centre of project management as they connect every other knowledge area. In our case, integration is done collectively by all members through our MSF approach, as no single leader is responsible; rather, all members contribute to maintaining the system's smooth integration across all phases. Our key integration points are the system block diagram, electrical schematics and classroom receiver hub design, as they serve as a shared reference point that defines how all subsystems interface with one another, ensuring that all members are aligned and working towards the same technical specification. Without active integration, subsystem conflicts and mismatches are very likely. For instance, firmware developed by one member might not align with the sensor wiring designed by another, or the power budget might not account for the full system load during transmission, which is why the block diagram is treated as the blueprint for the whole project. Moreover, the critical path analysis serves as the integration readiness checkpoint where all subsystems are to be verified independently before our software and hardware integration, which is our most important milestone. Through the action register and weekly meetings, the team strives to continuously monitor progress across all sectors, ensuring scope, schedule and technical decisions remain aligned so the final system is delivered as a unified, coherent and functional product rather than a collection of isolated components.